

NAS INTELLIGENCE PLATFORM

AI Work Portal

A NAS-centric multimodal AI workspace for business

Unify NAS intake, meeting capture, speech transcription, document and image OCR, hybrid RAG, generative models, MCP tools, and LINE collaboration to turn company content into searchable, traceable, reusable knowledge.

Feature snapshot: 2026-09-18

7+ Supported file classes	Local / Cloud Model execution choice	Hybrid RAG Vector + keyword retrieval	ZH-Hant / EN Bilingual interface
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All persistent information remains on the NAS

Source files, media segments, transcripts, translations, OCR text, PDF page previews, RAG chunks, vectors, model inputs and outputs, caches, and audit records are retained by the NAS. Content is sent externally only when a user explicitly selects a cloud model or external MCP service.

NAS Data Ownership - Keep source and derived assets in one lifecycle with permissions, reprocessing, download, and traceability.	Automated Intake - Watch the NAS inbox, browser recording, and uploads; notify users and enqueue new content immediately.
Local-first AI - Run LLM, embedding, ASR, OCR, and YOLO on the NAS, or select company-managed cloud APIs by policy.	Enterprise Governance - Centrally manage roles, the system LLM, LINE group policies, MCP allowlists, and model-call audits.

From Intake to Knowledge Asset

One NAS pipeline covers ingestion, source preservation, model processing, indexing, and downstream use.

01 Intake Recording, upload, NAS inbox, YouTube	02 Archive Source file and provenance metadata	03 Analyze ASR, OCR, YOLO, document parsing	04 Index Chunks, vectors, pages, previews	05 Use RAG, download, LINE, MCP
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Meetings and Audio - Browser recording and audio upload share the same ASR pipeline. - Local whisper.cpp, faster-whisper, SenseVoiceSmall, and Paraformer-zh, plus cloud ASR choices. - Large recordings are segmented and transcribed by background workers with visible progress, playback, and per-segment download. - Download or reprocess transcripts and normalize Chinese output to Traditional Chinese with OpenCC.	PDF, DOCX, and Images - Render every PDF page with PyMuPDF; extract text with pypdf and use PaddleOCR for scanned pages. - Image OCR supports JPG, PNG, WebP, TIFF, and BMP; DOCX parsing uses python-docx. - Chunks retain page_number, chunk_type, image_path, and page preview metadata. - Ask source-grounded document questions with the centrally selected system LLM.
Video Analysis - Ingest uploaded or YouTube video while preserving the original on the NAS. - Automatically segment large video for playback, download, and reprocessing. - YOLO defaults to CPU and falls back to CPU if requested accelerator inference fails. - Model choices include YOLO, vision-language runtimes, and cloud video interfaces.	Asset Operations - Browse audio, video, documents, and images from a unified NAS asset overview. - Retain processing model, time, status, error, and stage-by-stage results. - Download source files, media segments, and transcripts. - Requeue existing audio, video, PDF, DOCX, and image assets for processing.

Large-file processing

Browser uploads stream to storage with real progress and a 2 GB application default limit. Files above 100 MB, audio longer than 30 minutes, or video longer than 10 minutes can be segmented automatically; thresholds and segment lengths are deployment-configurable.

RAG, Models, and Cost Control

Search NAS retrieval and caches before generation so existing answers can be reused and repeated cloud API costs reduced.

01 Question User Prompt / System Prompt	02 Local Precheck Exact, keyword, similar questions	03 Hybrid RAG 75% vector + 25% keyword	04 LLM Local NAS or company cloud API	05 Persist Answer, sources, tokens, audit
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Qwen3 Embedding and Hybrid Retrieval	Semantic Cache and Autocomplete
- Qwen3-Embedding-0.6B defaults to the NAS loopback service at 127.0.0.1:8081. - Create vectors for new chunks and backfill existing chunks in the background. - Fall back to keyword retrieval when the embedding service is unavailable. - Return source file, page, scores, and image previews with answers.	- LLM and LINE queries support exact, semantic, and keyword-aware cache reuse. - Show similar historical questions and NAS cache state while the user types. - System Prompts use separate cache namespaces to prevent cross-instruction reuse. - Force Send bypasses cache and performs a real model call.

One System LLM	LLM -> MCP -> LLM
- Administrators select one current system LLM for AI Work, document RAG, translation, and LINE. - Use local Qwen3 4B / llama.cpp or a catalog of cloud APIs. - Separate User Prompt and optional System Prompt; send the latter using the provider's native system role. - Company API keys stay in the server environment and are never returned to the browser or database.	- The current LLM selects an authorized read-only MCP tool, retrieves data, then writes the final answer. - Built-in NAS MCP tools expose system status, shares, files, and recent activity. - Configure Monday.com bearer token and Linear read-only access; enable Google, Odoo, and others when authorized. - Persist tool input, output, caller, timestamp, and status in the NAS audit database.

Complete model-call records

For success, failure, policy blocks, and cache hits, retain timestamp, caller, provider, model, input, output, status, token or balance data, and source channel.

Collaboration, Governance, and Deployment

Apply company policy to people, models, external groups, and tool connectors with the NAS as a unified AI control plane.

Accounts and Access	LINE Enterprise Management
- Authenticated access with administrator and standard-user roles. - Create, disable, reset, and remove unused accounts. - Permission or password changes revoke existing sessions. - Model, MCP, LINE, and account management are administrator-only.	- Only administrator-approved company groups can use NAS, RAG, and company models; personal chats are always blocked. - Save, index, and summarize group PDF uploads with page citations. - Query authorized PDFs and meeting transcripts; push meeting summaries or full transcripts after completion. - Control monthly calls, token quotas, content versions, and cache invalidation.
Model and MCP Management	NAS Operations
- Show local model installation, size, download progress, path, and runtime readiness. - Download, cancel, retry, and validate added local models. - Manage MCP endpoint, transport, authentication environment variable, tool sync, and enablement centrally. - External tokens are not stored in the registry; secrets come from the NAS server environment.	- Retain source and derived data with SMB, NFS, and WebDAV intake scenarios. - Use WebSocket notifications for new files, processing state, and LINE jobs. - Bind local LLM and embedding endpoints to loopback only. - Keep traceable failure states and re-run processing from the asset detail page.

Reference deployment components

Layer	Component	Purpose
Portal / API	FastAPI + HTML / CSS / JavaScript	Authentication, upload, query, management, audit
Data and Files	NAS filesystem + SQLite	Assets, derivatives, chunks, vectors, policy
Local Generation	llama.cpp / Qwen3 4B	Default loopback 127.0.0.1:8080
Embedding	Qwen3-Embedding-0.6B	Default loopback 127.0.0.1:8081
Media and Docs	ffmpeg, ASR, PyMuPDF, PaddleOCR, YOLO	Enabled per installed runtime and NAS hardware

Deployment considerations

A visible feature does not imply that an external service is already authorized. Cloud LLM, cloud ASR, LINE, Google Workspace, Monday.com, Linear, Odoo, and other MCP services require the applicable API key, OAuth flow, bearer token, endpoint, or local runtime. Actual performance depends on NAS CPU, GPU, memory, storage, and network capacity.